Stylometric-Transformer Hybrid AI Text Detector

Key Facts, Architecture, Metrics, and System Reference Document

1. Project Overview

The **Stylometric-Transformer Hybrid AI Text Detector** is an advanced Natural Language Processing system designed to distinguish human-authored text from text generated by Large Language Models (LLMs like ChatGPT). It achieves high classification accuracy and interpretability by combining the deep semantic representation power of **RoBERTa contextual embeddings** with **16 hand-crafted linguistic stylometric features**.

2. Dataset

The project is trained and evaluated on the **HC3 (Human ChatGPT Comparison Corpus)** dataset. The corpus was rigorously preprocessed, cleaned, balanced across classes (50% Human / 50% AI), and split into training, validation, and test subsets. It covers a diverse set of real-world domains including Open QA, Reddit ELI5, Wikipedia Q&A;, Medicine, Finance, and Law.

3. Feature Engineering

The system extracts 16 quantitative stylometric features spanning 4 core linguistic dimensions: sentence structure, lexical diversity, part-of-speech (POS) distributions via spaCy, and readability/punctuation metrics. Each feature was statistically validated using **Welch's t-test** and **Cohen's d** effect size:

- **Top AI Discriminator (Vocabulary Diversity):** Type-Token Ratio (TTR) exhibited the largest positive separation (**Cohen's d = 1.53**), reflecting higher lexical variety and broader vocabulary choice in AI text.
- **Top Human Discriminator (Sentence Length):** Average Sentence Length exhibited a strong negative effect (**Cohen's d = -0.84**), reflecting that human writing frequently features longer, compound-complex sentences compared to ChatGPT's consistent, moderate sentence lengths.
- **Additional Features:** Sentence length standard deviation (burstiness), POS ratios (noun, verb, adjective, adverb, pronoun), punctuation frequencies (commas, periods, question marks), paragraph length, and readability scores (Flesch-Kincaid Grade Level, Flesch Reading Ease).

4. Model Architecture

The system compares a fine-tuned **RoBERTa baseline model** against a specialized **HybridClassifier** neural network:

- **Transformer Semantic Branch:** Passes input tokens through RoBERTa-base to extract the 768-dimensional contextual pooling representation (from the [CLS] token).
- **Stylometric Feature Branch:** Normalizes the 16 features via a fitted StandardScaler and passes them through a dense feedforward network (16 $\rightarrow$ 128 hidden units with ReLU activation and Dropout = 0.2).
- **Fusion Head:** Concatenates both representations into an 896-dimensional multimodal vector, followed by a classification head (896 $\rightarrow$ 256 $\rightarrow$ 2 classes with Dropout = 0.3) yielding calibrated probabilities.

5. Results & Evaluation Metrics

Benchmarked on the held-out HC3 test split (evaluated against ground-truth labels):

Model Architecture	Accuracy	Precision (AI)	Recall (AI)	F1-Score (AI)
RoBERTa Baseline	95.47%	94.85%	96.20%	95.52%
Hybrid Model (RoBERTa + Stylometrics)	**96.79%**	**96.14%**	**97.48%**	**96.81%**

6. Pipeline & Codebase Structure

File / Path	Component Purpose & Implementation
`predict.py`	Core end-to-end inference engine. Loads hybrid_model.pt, RoBERTa tokenizer, and scaler.pkl. Extracts 16 features on the fly with spaCy, scales them, and executes the forward pass.
`predict_long.py`	Chunked wrapper for long text (>150 words). Splits text at sentence boundaries into ~150-word chunks, runs inference on each, and returns a word-count-weighted average AI probability. Also extracts unscaled full-text features for explanation.
`explain.py`	Feature-level interpretability engine. Computes directional pull (Human vs. AI) by comparing the text's 16 stylometric features against precomputed class training baselines (feature_stats.json) using Cohen's d weighting.
`app/streamlit_app.py`	Interactive Streamlit web application. Provides paste-text input, multi-page PDF document upload, live sample presets, result cards, Plotly probability gauge, chunk-by-chunk probability progression, and an expandable 16-feature diagnostic drawer.
`saved_models/scaler.pkl`	StandardScaler fitted on the training split across all 16 stylometric features in exact canonical column order.
`saved_models/hybrid_model.pt`	Trained PyTorch state dictionary checkpoint for the HybridClassifier architecture.
`saved_models/feature_stats.json`	Precomputed feature mean and standard deviation benchmarks for Human and AI classes used by explain.py.
`analysis/`	Suite of benchmarking and evaluation scripts: evaluate_models.py (test evaluation), plot_comparison.py (confusion matrices / charts), feature_distributions.py (statistical analysis), and compute_feature_stats.py.

7. Streamlit Web Application Features

- **Dual Ingestion Modes:** Supports direct manual text input with word/character counters or multi-page PDF document upload.
- **PDF Processing:** Powered by pypdf with automatic page extraction, layout-aware text joining, and a proactive warning for scanned or image-only documents (<20 extracted words).
- **Visual Verdict Cards:** High-contrast classification verdict displaying final label (Human-written or AI-generated) with exact confidence percentages.
- **Plotly Probability Gauge:** Dynamic, color-coded dual probability distribution bar visualization.
- **Chunk-Level Breakdown:** For long texts (>150 words), presents chunk-by-chunk sentence segment cards with individual AI probabilities and length metrics.
- **Explainability Diagnostic Drawer:** Interactive panel showing top-8 feature pull chart (directional shift toward Human vs. AI) and a comprehensive 16-feature comparison table against dataset norms.

8. Known Limitation & Diagnostic Findings

During verification against an isolated diagnostic batch of 25 real AI-generated samples from the HC3 evaluation split, the model achieved **92.00% recall (23/25 correctly identified)**. This matched its exact performance on those same 25 samples in the official offline evaluation (which achieved 97.69% AI recall over the full 476 AI test samples):

- **Analysis of Misclassifications:** The 2 misclassified samples (AI samples predicted as Human) were formal, encyclopedic entries (e.g., historical sitcom summaries and sex education definitions). Both landed at borderline probabilities (~48.2% and 49.0% AI probability).
- **Root Cause:** This is an expected behavioral phenomenon of stylometric convergence — highly polished, factual expository text written by LLMs naturally aligns with the sentence length variation and syntactic regularity of formal human reference texts. It is not an artifact of code degradation.
- **Integrity Verified:** Feature order, scaler parameters (scaler.pkl), and model weights (hybrid_model.pt) were verified to be strictly identical between training, evaluation, and live app inference.

9. Tech Stack Summary

Core Runtime & Frameworks: Python 3.10+, PyTorch, HuggingFace Transformers (roberta-base), spaCy (en_core_web_sm), scikit-learn, textstat, SciPy, joblib.
User Interface & Ingestion: Streamlit, Plotly Graph Objects, pypdf.
Data Science & Evaluation: Pandas, NumPy, Matplotlib, Seaborn, HuggingFace Datasets.